Viroopaksh Chekuri

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Education

University of California San Diego, M.S. Computer Science (AI), *San Diego, CA* | **GPA: 3.97/4.0** | Sep 2024 – Jun 2026
Indian Institute of Technology Madras, B.S. Data Science & Applications, *Chennai, TN* | **GPA: 3.87/4.0** | Jan 2021 – Dec 2024
Indian Institute of Technology Madras, B.Tech. Metallurgical & Materials Engineering, *Chennai, TN* | **GPA: 3.50/4.0** | Jul 2019 – Jul 2023

Relevant Experience

AI Engineer Intern, Artemis Life Sciences **San Diego, CA, USA** **Jul 2025 – Sep 2025**

- **Eliminated manual regulatory document aggregation bottlenecks** by architecting a **scalable end-to-end ETL pipeline** that scraped FDA open databases via **Selenium**, structured outputs to **JSON**, and uploaded **100K+ medical device JSONs** and **3K+ device summary PDFs** into **AWS S3**.
- **Solved unstructured regulatory data retrieval challenges** by implementing structure-aware JSON/PDF chunking pipelines using LangChain and OpenAI embeddings, vectorizing content and indexing into Pinecone for semantic search.
- **Designed and deployed a hybrid Retrieval-Augmented Generation (RAG) system** using GPT-4o with hybrid search (dense + structured retrieval), enabling health consultants to summarize and query regulatory data with near-complete automation, reducing manual search effort by **~100%**.
- **Built a ReAct-based AI agent using LangGraph** capable of drafting approval documents by dynamically retrieving context from conversation memory, vector databases, and predefined templates, automating domain-specific document generation workflows.
- **Engineered production-grade AI services** emphasizing modular ingestion, retrieval transparency, and maintainable agent orchestration pipelines suitable for enterprise regulatory environments.

Research Assistant, CAIDA, SDSC **La Jolla, CA, USA** **May 2025 – Jul 2025**

- Engineered **LLM-driven** literature **extraction pipelines** with **retrieval augmentation** and **structured output enforcement**, converting unstructured scientific papers into normalized research datasets while **reducing manual keyword lookup** and **context matching** by **50%**.
- Designed a **multi-LLM evaluation** and ranking framework (**LLaMA3, Gemma, DeepSeek**) to validate whether extracted keywords were truly referenced and research-relevant, improving **precision** of **structured outputs**.
- Benchmarked **foundation models** across **cost**, **latency**, and **accuracy** using reproducible evaluation workflows with **automated regression testing** and **metric tracking** to prevent **performance degradation** across prompt iterations.

Machine Learning Researcher, IIT Madras **Chennai, TN, IN** **Apr 2023 – Sep 2023**

- **Led a 5-member research team** to design an end-to-end Mridangam single-stroke classification system within an 11-week research cycle, translating signal processing theory into **deployable ML models**.
- **Solved low-level audio signal discrimination challenges** by engineering **rolling-window time-series statistics** and spectral distribution features from a **6,977-file MIRDATA Mridangam dataset**, extracting discriminative frequency-domain representations.
- **Built an interpretable Random Forest classifier** to balance accuracy and overfitting risk, achieving **90.3% classification accuracy** while preserving model explainability for research validation.
- **Achieved signal-processing-based state-of-the-art performance** in Mridangam single-stroke classification, improving stroke-level precision across multiple percussive classes.
- **Established reproducible experimental workflows** for feature engineering and model evaluation, contributing to a broader real-time transcription research initiative in Carnatic percussion music

NLP Engineer Intern, Willings Inc. **(Remote) Tokyo, JPN** **Jun 2022 – Jul 2022**

- **Addressed inefficiency in recruiter resume screening** by designing and deploying a lightweight resume categorization system optimized for constrained compute environments; fine-tuned a Softmax classifier on top of frozen BERT embeddings derived from structured NER entities, achieving **85% classification accuracy** using a minimal training dataset.
- **Solved resume format inconsistency (PDF, DOCX, image-based resumes)** by engineering a structured information extraction pipeline using spaCy-based custom NER models, enhanced tokenization logic, PDF loaders, and OCR APIs to standardize multi-format resume ingestion.
- **Designed a production-ready REST API service** to serve the trained model for real-time inference, enabling seamless integration with recruiter workflows and reducing resume grouping time by **~80%**.
- **Improved model generalization under data scarcity constraints** by structuring NER-extracted entities into semantically meaningful strings before classification, maximizing signal from limited labeled data.
- **Demonstrated strong applied ML competency beyond internship scope** by building an interpretable XGBoost model in the AMEX Credit Default Detection competition (50K rows, 191 features), achieving **~0.90 precision** on fraud prediction through feature engineering and imbalance-aware training.

Projects

- **Industrial Predictive Maintenance System with ONNX Deployment:** Designed and trained **PyTorch**-based sequence models (**LSTM, Temporal CNN**) to predict **Remaining Useful Life** (RUL) on industrial sensor **time-series** data. Engineered **preprocessing pipelines** for **multi-sensor degradation signals** and implemented **sliding-window feature generation**. Exported trained models to **ONNX** and deployed inference using **ONNX Runtime (C++)**, achieving low-latency **CPU inference**. Benchmarked **ONNX** vs native **PyTorch** inference performance and memory footprint. Developed model evaluation framework with RMSE, **degradation trend visualization**, and early-failure detection analysis.
- **Reinforcement Learning-Based Industrial Process Controller:** Developed **reinforcement learning controller** (PPO/DDPG) in **PyTorch** to optimize simulated industrial process parameters under **dynamic constraints**. Designed **custom Gym environment** modeling process temperature and pressure dynamics with **reward shaping for yield optimization**. Trained **policy networks** for **stable control** and exported trained models to **ONNX** for **low-latency deployment**. Evaluated controller robustness under **noise** and **system perturbations**, demonstrating adaptive **optimization** behavior.

Skills

- **Frameworks & Tools**:PyTorch, ONNX Export & Optimization, ONNX Runtime, HuggingFace Transformers, FastAPI , PEFT, BitsAndBytes, FAISS, LangChain, AWS, GCP, Flask, Streamlit, Scikit-learn
- **Generative AI & Systems:** LLMs, RLHF, QLoRA/LoRA Fine-Tuning, Agentic AI Systems, Vector Databases(FAISS, Pinecone), Model-Context Protocol (MCP), LangChain, Distributed AI, RAG Architectures, AI Pipelines, AI Model Optimization and Deployment, Tool-Augmented Reasoning, Inference Efficiency
- **ML Production Systems:** MLflow, Docker, Shadow Deployment, Drift Detection, Airflow, Apache Kafka, PySpark, Great Expectations, Jenkins
- **Agentic AI & LLM Systems:** Multi-Agent, Tool-Oriented AI Systems, Hallucination Mitigation, LLM Routing & Refusal, Cost Governance, FastAPI
- **Systems and Infrastructure:** PostgreSQL, FastAPI, Async Workers, Microservices, GPU Optimization, Real-time Inference, Distributed Processing.